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Executive Summary

Research Contributions

My research centers on movement generation and analysis at the intersection of neuroscience, biomechanics, and AI. I developed the NEUROiD spino-musculoskeletal co-simulation platform, enabling in-silico modeling of upper and lower limb movement, spasticity progression, and reinforcement learning-based motor control. This work lays the foundation for digital twins and virtual patients. I also established a state-of-the-art Human Movement Analytics Lab with synchronized MoCAP, EMG, gaze tracking, and mobile camera pipelines, now commercialized via Bheri Intelligence and Analytics Pvt Ltd. My contributions extend to generative AI for bidirectional movement-text transformation and heritage visualization systems. My Publications include 13 Journals, 9 conferences, 5 book chapters, 1 granted Patent, 1 Monograph and 1 Book.

Grants secured

I have secured over ₹7.5 Crore in competitive research funding as Principal Investigator, including major grants from DST, MEITY, SERB, and MoE. These support projects in neurobiomechanics, spinal cord modeling, movement disorder simulation, and Heritage Sciences.

Teaching & Mentoring

Over the past decade, I have taught **132 course credits** across **16 unique courses**, including foundational and advanced topics in neuroscience, AI, movement science, and Heritage Sciences. I am mentoring/ have mentored 15 PhD students of whom 5 have graduated/completed OC.

Department and Institute Development

I co-founded the Centre for Healthcare Entrepreneurship (CfHE), securing \$4M in funding and mentoring its first five startups. I have mentored multiple undergraduate and postgraduate batches, coordinated placements (2014–25), and served on the DPGC-BME (2018–23). My design of basket-based B.Tech curriculum in Biomedical Engineering has been nationally replicated. As Founding Head of the Department of Heritage Science & Technology, I launched India's first M.Tech and PhD programs in Heritage Sciences, developed multidisciplinary lab clusters under the Takshashila Centre of Excellence, and authored foundational IKS articles, monograph and a textbook. These efforts earned me the Adani Chair in Indology (2024–29) and positioned IIT Hyderabad as the national hub for Heritage Science and Indian Knowledge Systems.

Grants

<i>Granting agency</i>	<i>Grant amount (lacs)</i>	<i>Title of Project</i>	<i>Duration</i>
As Principal Investigator			
MoE SPARC	91.6	Design and validation of upper limb orthopedic implant using closed loop neurobiomechanics simulation	2023-25
DST	527	Takshashila Centre for multidisciplinary Heritage Science and Technology	2022-25
DST	48.05	Reconstructing spinal cord neural dynamics during exercise and yoga	2020-23
MEITY	31.25	Neuroinformatics and simulation platform for evaluation of movement disorders	2019-22
SERB	24.1	Development of a reusable model component library for human spinal cord modeling and simulation	2015-18
As Co-investigator			
BIRAC	82.8	Indo-Italy networks of Excellence (IITH, IITM, Amrita University)	2022-25
BIRAC	42.7	Surgeo Assist: An indigenous neurosurgical assistance platform for safer spinal surgeries	2020-21

Awards

Adani Indology Chair (2024-29)

Publications - Journal

Srivastava, A., Sridharan, K.S., **Raghavan, M.**, (2025) Pose Estimation in Indian Sculptures Using Hierarchical Assembly of Keypoints. *ACM Journal on Computing and Cultural Heritage*, (**Accepted**)

Singh, A.K., Sridharan, K.S., Wu, S., Tirosh, O. and **Raghavan, M.**, (2025), In-silico neuro-musculoskeletal model demonstrates spasticity progression with descending motor tracts loss in simulated clinical triage. *Computers in Biology and Medicine*, 192, p.110270.

Ganguly, S., Chhaya, M.M., Jain, A., Koppula, A., **Raghavan, M.** and Sridharan, K.S., (2024). Mark3D—A semi-automated open-source toolbox for 3D head-surface reconstruction and electrode position registration using a smartphone camera video. *Medical & Biological Engineering & Computing*, pp.1-13.

Rangayyan, Y.M., Kidambi, S. and **Raghavan, M.**, (2023). Deaths from undetected COVID-19 infections as a fraction of COVID-19 deaths can be used for early detection of an upcoming epidemic wave. *Plos one*, 18(3), p.e0283081.

Kapardi, M., Pithapuram, M.V., Rangayyan, Y.M., Iyengar, R.S., Singh, A.K., Sripada, S. and **Raghavan, M.**, (2022). In-silico neuro musculoskeletal model reproduces the movement types obtained by spinal micro stimulation. *Computer Methods and Programs in Biomedicine*, p.106804.

Pithapuram, M.V. and **Raghavan, M.**, (2022). Automatic rule-based generation of spinal cord connectome model for a neuro-musculoskeletal limb in-silico. *IOP SciNotes*, 3(1), p.014001.

Mahankali, N.S., **Raghavan, M.** and Channappayya, S.S., (2021). No-Reference Video Quality Assessment Using Voxel-wise fMRI Models of the Visual Cortex. *IEEE Signal Processing Letters*.

Raghavan, M. and Maigur, R.K., (2021). Indian Costumes through the Ages: Insights into Sartorial Choices Using Mathematical Biology. *Archaeometry*. <https://doi.org/10.1111/arcm.12681>

Raghavan, M., Sridharan, K. S., & Mandayam Rangayyan, Y. (2020). Using epidemic simulators for monitoring an ongoing epidemic. *Scientific reports*, 10(1), 16571. <https://doi.org/10.1038/s41598-020-73308-5>

Iyengar, R. S., Pithapuram, M V., Singh, A K., **Raghavan, M.** (2019) 'Curated model development using NEUROiD: A web-based NEUROmotor integration and Design platform', *Frontiers in neuroinformatics*. *Frontiers*, 13, p. 56.

Sankaranarayani, R., **Raghavan, M.**, Nalini, A., Laxmi, T.R. and Raju, T.R., (2014). 'Reach task-associated excitatory overdrive of motor cortical neurons following infusion with ALS-CSF'. *Journal of Neural Transmission*, 121(1), pp.49-58.

Raghavan, M., Amrutur, B., Narayanan, R. and Sikdar, S.K., (2013). 'Synconset waves and chains:

spiking onsets in synchronous populations predict and are predicted by network structure'. *PloS one*, 8(10), p.e74910.

Raghavan, M., Amrutur, B., Srinivas, K.V. and Sikdar, S.K. (2012) 'A study of epileptogenic network structures in rat hippocampal cultures using first spike latencies during synchronization events', *Physical biology*. IOP Publishing, 9(5), p. 56002.

Publications – Conference

Das, A. and **Raghavan, M.**: A Comparative Assessment of Open-source Building Footprint Data for Indian Heritage Sites: Towards a Shape-preserving 3D mapping, *ISPRS Ann. Photogramm. Remote Sens. Spatial Inf. Sci.*, X-M-2-2025, 73–79, <https://doi.org/10.5194/isprs-annals-X-M-2-2025-73-2025>, 2025.

Krishna, S.H., Chinthapenta, V., **Raghavan M.**, Unni., H.N., (2024)., Finite Element Modelling Of an Anatomically Accurate Human Spinal Cord. In *Proceedings of the 10th World Congress on Electrical Engineering and Computer Systems and Sciences (EECSS'24)* DOI:10.11159/icbes24.110

Mallampalli, K., Sarathy, K., Balasubramanian, R. and **Raghavan, M.**, (2024), July. Studying spinal feedback loops post tendon transfer surgery using the pincer grasp task: A Neuro-Musculoskeletal study using NEUROiD. In *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)* (pp. 1-4). IEEE.

Singh, A.K., Sridharan, K.S., Wu, S., Tirosh, O. and **Raghavan, M.**, (2024), June. Simulated Clinical Triage of Modified Tardieu Test for Lower Limb Spasticity. In *2024 IEEE International Symposium on Medical Measurements and Applications (MeMeA)* (pp. 1-6). IEEE.

Rahaman, J., Jain, A., Sridharan, K.S. and **Raghavan, M.**, (2023), October. A Rule-based Semi-automated OCR Postprocessing Method for Aligning Multi-language Transcripts with Multi-column Text. In *2023 First International Conference on Advances in Electrical, Electronics and Computational Intelligence (ICAEECI)* (pp. 1-9). IEEE.

Iyengar, R.S. and **Raghavan, M.**, (2020), December. MPI Parallelization of NEUROiD Models Using Docker Swarm. In *2020 IEEE 26th International Conference on Parallel and Distributed Systems (ICPADS)* (pp. 655-660). IEEE.

Prakash, S.C., Ganguly, S., Yadav, P.K., **Raghavan, M.** and Sridharan, K.S., (2020, July). Evaluation of a gamified upper-arm bimanual trainer for stroke patients-A healthy cohort study. In 2020 International Conference on Signal Processing and Communications (SPCOM) (pp. 1-5). IEEE.

Mallampalli, K., Patel, S., Iyengar, R.S., Sridharan, K.S. and **Raghavan, M.**, (2020, July). Tool for image annotation based on gaze. In 2020 International Conference on Signal Processing and Communications (SPCOM) (pp. 1-5). IEEE.

Ramamurthy, S. and **Raghavan, M.** (2013) 'Filter design for synthesis of musical notes: A multidimensional feature-based approach', in 2013 IEEE International Conference on Signal and Image Processing Applications. IEEE, pp. 106–111.

Publications – Book Chapters

Singh, A.K, Mallampalli, K., Rangayyan, Y.M., Iyengar, R.S., **Raghavan, M.**, (2026), *NEUROiD: Building Neuro-musculoskeletal Models of Upper and Lower Limb In-Silico by Bridging Neural Circuitry, Biomechanics, and Intelligent Control*. In *Bridging the Gap between Mind and Machine*, Springer Nature, (**In Press**)

Raghavan, M., (2025), *Multidisciplinary IKS & Heritage: Areas of Study in an IKS-Inspired Academic System.*” *Samvādasamgrahaḥ Lecture Notes in IKS*, pp 30–66., Ministry of Education, Government of India, <https://doi.org/10.71412/iks0007.ch2>. [[Link](#)]

Raghavan, M., Simha M S H, and Ramaswamy C R. (2025) *Indic Knowledge Systems and Heritage: Elements & Design Principles*. *Samvādasamgrahaḥ: Lecture Notes in IKS*, pp 3–29., Ministry of Education, Government of India, <https://doi.org/10.71412/iks0007.ch1>.

Iyengar, R.S., Mallampalli, K., Singh, A.K., Koppula, A., Sridharan, K.S. and **Raghavan, M.**, (2023). The NEUROiD neuromusculoskeletal movement simulation platform. In *Digital Human Modeling and Medicine* (pp. 161-197). Academic Press.

Raghavan, M. ., and Maigur, R. K., (2021). *Pratyāhāra – A Neuroscientific perspective of Yoga Shastra*. *Yoga Vijnana* 1(1), p.56. MDNIY, Ministry of AYUSH.

Publications – Monograph

Medury, R.K., Kotak, B.P., Mandayam Rangayyan, Y., Sridharan, K.S., and **Raghavan, M.** “Indian Knowledge Systems & Heritage: State of the Industry Report,” (2024). Ministry of Education, Government of India. <https://doi.org/10.71412/cinu1473>. ISBN: 9788197811562

Publications – Book

Raghavan, M., Simha, M.S.H. and C.R., Ramaswamy. (2023). Elements of Indic Knowledge Systems and Heritage. HTSR Institute, ISBN 978-81-964176-0-4.

Patents

1. Granted Indian Patent No. 566402 (Application number 202441046360)

Raghavan, M., Tipnis, A.A., Sridharan, K.S.

Title: A METHOD AND A SYSTEM FOR REMOTELY MONITORING AN INTENSIVE CARE UNIT.

2. Indian Patent application no. 202441095905 (Published)

Raghavan, M., Tipnis, A.A., Koppula, A., Rangayyan, Y.M., Sridharan, K.S.

Title: A MULTIMODAL INTERACTION OR CONFERENCING SYSTEM FOR VIRTUAL ASSESSMENT AND A METHOD THEREOF

3. Indian Patent application no. 202441036947 (Published)

Sridharan, K.S., Ganguly, S., **Raghavan, M.**

Title: “A system for continuous estimation and feedback of arousal and valence states using measures of the cardiac cycle state

Posters & Meetings

Pithapuram, M. V., Iyengar, R. S. and **Raghavan, M.** (2019) ‘In silico Spinal cord model shows the viability of targeting segmental foci along rostrocaudal axis for eliciting a variety of movement types’, BMC Neuroscience, 28th Annual Computational Neuroscience Meeting: CNS*2019 in, 20(1), p. P254. doi: 10.1186/s12868-019-0538-0.

Singh, A. K., Pithapuram, M. V. and **Raghavan, M.** (2020) ‘Coupling neural simulations with inverse kinematics through proprioceptive afferents’, in Virt. Phys. Hum. Conference (**Accepted as Poster**), 2020.

Pithapuram, M. V., Iyengar, R S., Singh, A K., **Raghavan, M.** (2020) *A Neuromechanical model of lower limb with lumbosacral spinal cord controller for InSilico spinal cord stimulation experiments*, in *Virt. Phys. Hum. Conference (Accepted for Oral presentation)* , 2020.

Teaching

Course Code(s)	Course Name	Credits	# Times Taught	# Credits Delivered
BM1060, BM5050, HT5030, AI3002	Brain and Neuroscience / Introduction to Brain and Neuroscience	1	23	23
BM2033	Probability and Random Processes	2	4	8
BM2043	Algorithms and Data Structure Lab	2	1	2
BM3000	Foundation of Natural Intelligence	3	3	9
BM4025	Implementations of Natural Intelligence	2	3	6
BM4060	Movement Science	2	3	6
BM4061	Movement Science Lab	2	3	6
BM4081	Computational Neuroscience Lab	2	3	6
BM5030	Scientific Computing	1	5	5
BM6140	Theoretical and Computational Neuroscience	2	10	20
BM6150	Mathematical Physiology and Modeling	2	2	4
BM6243	Neuromechanics	2	8	16
BM6330	Theoretical & Computational Neuroscience	3	1	3
EE7710	Probabilistic Models of the Brain	3	1	3
HT5000	Elements of Indian Heritage	2	3	6
HT5053	Human Movement–Measurement and Control	1	1	1
HT5123	Digital Visualization	2	4	8
	Total Course Credits taught during Aug 2014 – Aug 2025			132

Summary of Contributions and Impact

Research Contributions

My research focuses on Movement Generation and Analysis and uses computational methods at the interface of Neuroscience, Biomechanics and Artificial Intelligence.

Movement generation is an outward process, where motor intent in the motor cortex is shaped into visible musculoskeletal movement via spinal cord circuits and modulated by Basal Ganglia and Cerebellum. Our primary contribution over the years has been the development of the NEUROiD Spino-Musculoskeletal co-simulation system that generates movement using NEURON and OpenSim simulators. We further used NEUROiD to create two exemplar model platforms for the upper and lower limb. In each of these we demonstrated movement generation that is validated by experiment results in health and disease. In the upper limb we simulated movement generation with and without a passive implant. In the lower limb we explored the emergence of spasticity and clinical triaging for the same. We also probed the progression of learning in the exemplar upper limb system under the action of a goal directed Reinforcement Learning System such as the Basal Ganglia. Our work in biophysical movement generation lays a solid platform for development of virtual patients and digital biological twins.

Movement analysis is an inward process starting from observed movement in physical space and tries to deduce latent biomechanical and neurological parameters. Human Movement Analysis has significant applications in real world ranging from physiotherapy, dance, sports, martial arts to security and defense. Our primary contribution in this area has been to establish a SOTA facility for human movement analytics that includes MoCAP, EMG, Force Plates, Gaze Tracking synchronized with mobile cameras and an optical synchronizer. A contribution of high utility has been the comprehensive study of the performance of mobile camera based Human Pose estimation in comparison to the SOTA MoCAP. Our lab has developed significant knowhow in developing camera based gait analysis pipelines leading to strong interest from the industry and armed forces. This work has since grown into a startup – Bheri Intelligence and Analytics Pvt Ltd, incubated at iTTC.

While movement analysis and generation approach the movement pipeline from opposite ends, it is hoped that they will meet in the coming decade, allowing development of movement generating digital twins that are customized using motion analytics. In the near term, generative AI offers a viable means to bridge movement intent with actual movement, though shorn of the biophysical mechanisms. Our lab has started early work on mastering the generative AI pipelines to perform a bidirectional Text (Movement Description) - Movement transformation. In the decade to come it may be possible to use this technology to create a fully customizable digital twin.

Academic Development

As the Founding Co-Head (along with Prof. Renu John) of the Centre for Healthcare Entrepreneurship (CfHE) from 2015 to 2018, I co-led the investor pitch that secured \$4 million

in funding, designed and coordinated the flagship Fellowship Programs, and mentored the first five startups incubated at CfHE—Nemo.Care, BeAble Health, Kvayat Medical, Heamac Healthcare, and Jivika Healthcare. I designed and initiated the B.Tech program in Biomedical Engineering, overseeing curriculum development, regulatory approvals, placement strategy, and student mentorship through graduation. The program's distinctive basket-based curriculum focusing on contemporary areas such as Medical Devices, Biomechanics, Computational Neuroscience, Tissue Engineering has since been replicated across institutions nationwide. I also co-organised (along with Dr. Kousik Sarathy) the redesign of the M.Tech BME programs, introducing streamlined academic tracks that significantly improved placement outcomes. I have served as Placement Coordinator for Biomedical Engineering from 2014 to 2025, supporting student career development and industry engagement. Additionally, I have been Faculty Advisor for multiple batches across B.Tech-BME, M.Tech-AI, M.Tech-BME, and M.Tech-HST, and served on the Departmental Postgraduate Committee (DPGC-BME) from 2018 to 2023.

Institute development

As the Founding Head of the Department of Heritage Science & Technology at IIT Hyderabad (2022–25), I initiated India's first M.Tech and PhD programs in Heritage Sciences, introducing specialisations such as Yoga Technology, Indic Language Processing, Conservation, and Archaeometry. I led the development of a multidisciplinary lab cluster under the Takshashila Centre of Excellence, including computing infrastructure for Heritage Data Sciences and Gen-AI research. My invited articles and Monographs on IKS & Heritage have highlighted the economic potential of Heritage Sciences and is used as a reference document for building a systematic framework linking academic research to industrial activity. My book on the “Elements of Indic Knowledge Systems and Heritage” is now considered as one of the primary resource materials for undergraduate and graduate studies by the IKS Division. These efforts have seeded platform technologies for public good, now being incubated under our Section-8 startup, Bheri Technology Foundation.

My work in Indian Knowledge Systems and Heritage research was recognized by the award of Adani Chair in Indology (2024–29). Collectively, these initiatives have positioned IIT Hyderabad as the national leader in rigorous, scientific approaches to Heritage Sciences. This recognition is reflected in my national roles, including representing the Ministry of Education's IKS Division in curriculum development and capacity building across Indian universities (2022–25), leading rural development initiatives with the Ministry of Tourism, chairing the independent committee to review the IKS Division (Jan 2024), contributing to the design and rollout of the Multidisciplinary Minors Program in IKS (Sep 2024), and authoring a national report on the state of India's Heritage industry.

Education

- Post-Doctoral Fellow at National Centre for Biological Sciences-TIFR, Bangalore (2014)
 - Vatsala Lab; Motor Systems Modeling
- Interdisciplinary PhD in Computational Neuroscience at IISc, Bangalore (2008 – 2014)
 - First Spike Latency based coding in neuronal networks
- All-India-Rank 12 in Graduates Aptitude Test in Engineering (GATE- 2008)
 - GATE - Computer Science – 2008
- 7 years of professional experience at Motorola India (2001-2008)
 - Embedded systems, DSP and device drivers.
 - Part of global team that created the world's first 3G phone
- “Bachelor of Engineering” in Computer Science and Engineering from Bangalore University